

Foundation Models Used Acknowledgment

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Unit: FIT2099 S1 2026

Assignment: Assignment 1 - Garbage Collection Inc.

Foundation Models Used: ChatGPT / Claude

Specific Contributions and Guided Usage

I used the foundation models mentioned above in a guided fashion to assist with the following technical components of the assignment:

1. Requirement 2: Fire Stacking Logic (FireStackable & Fire Ground)

Purpose: Guidance on managing overlapping status effects on a single tile.

AI Contribution: The model assisted in designing a robust logic for the Fire ground class to handle multiple "stacks" of fire originating from different lantern leaks. It suggested using a collection (List<Integer>) to track the remaining turns of each stack independently, ensuring that damage scales correctly as clarified in Ed Discussion threads.

2. Requirement 4: Facility Alarm System (AlarmManager)

Purpose: Implementation of a global state manager for the facility-wide alarm.

AI Contribution: I consulted the model for best practices in implementing the Singleton Pattern and the Observer Pattern within the FIT2099 engine framework. The AI provided guidance on how the AlarmManager should maintain a list of AlarmListener subscribers and notify them during state transitions (Activation/Deactivation).

3. Requirement 4: Pathfinding Logic (HuntBehaviour)

Purpose: Algorithm for hostile actors to pursue the nearest worker.

AI Contribution: The model helped guide the implementation of the findNearestWorker method. Specifically, it provided the logic for iterating through the GameMap efficiently to identify conscious actors with the WORKER ability and calculating the Manhattan Distance between locations to determine the optimal movement exit for the operate method.

Acknowledgment of Responsibility

I acknowledge that while foundation models were used to guide the implementation of these specific logic blocks, I am entirely responsible for the final code submitted. I have verified all citations and code logic for accuracy.